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Phased Antenna Array Optimization using CST and Python

Alexander Volkov

Abstract – The paper shows a simple example of phased antenna array optimization using CST and Python. Sometimes it is required to provide minimum average S_{11} though array scanning range. In this case the optimization should be carry out at different scanning angles. The example, presented in this paper, shows one of the possible ways to execute it.

Keywords – Phase antenna array, CST, Python, automation, optimization, VSWR, scanning range, co-simulation.

Using Python in CST

The `cst` package has been added to CST Studio Suit 2020 in order to provide a Python interface to the CST Studio Suite. It contains the three packages:

cst.interface – Allows controlling a running CST Studio Suite.

cst.results – Provides access to 0D/1D Results of a `cst` file.

cst.eda – Provides an interface to a Printed Circuit Board (PCB).

The `cst` package can be used by designers to interact with CST using all Python advantages and libraries.

In this paper, my initial experience of using the `cst` package is presented. I decided to start my studying by using it for optimization of a simple phased antenna array. Other programming languages, such as MATLAB with ActiveX [1-3] or special libraries [4] or VBA, can be used to solve this problem.

Design Example

Linear dipole phased antenna array is used as an example for the optimization using CST and Python. The array element, shown on Fig. 1, is designed in CST. It has the following parameters: dl – dipole length (*will be used for optimization*), $dw = 2$ mm – dipole width, $dg = 2$ mm – gap between dipole arms, h – distance between the ground plane and the dipole (*will be used for optimization*), $size_y = 100$ mm – ground plane length in y -direction, $size_x = 65$ mm – array step, $t = 0$ mm – dipole thickness, $t_{pec} = 0$ mm – ground plane thickness, θ – beam direction. Fig. 2 depicts the setting of the project, such as

boundary conditions (Fig. 2 a, b), mesh properties (Fig. 2 c) and solver parameters (Fig. 2 d).

The antenna array has to have the frequency operation bandwidth from 2.9 GHz to 3.1 GHz, the scanning range from -45° to 45° and has a minimum average S11 through its operation bandwidth and scanning range.

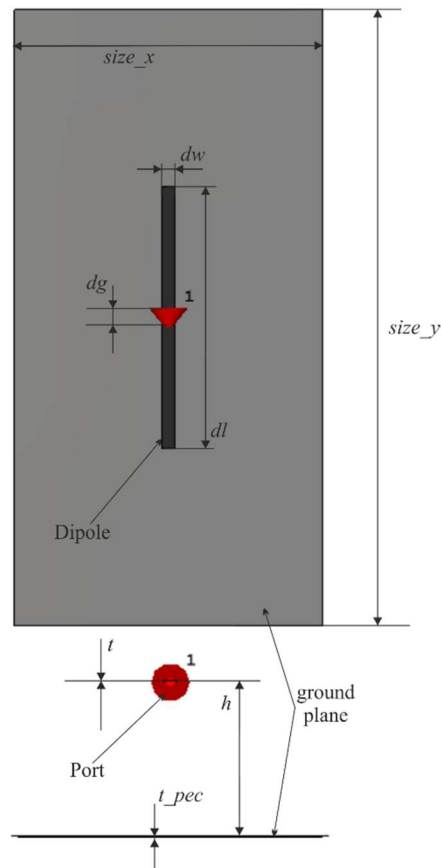
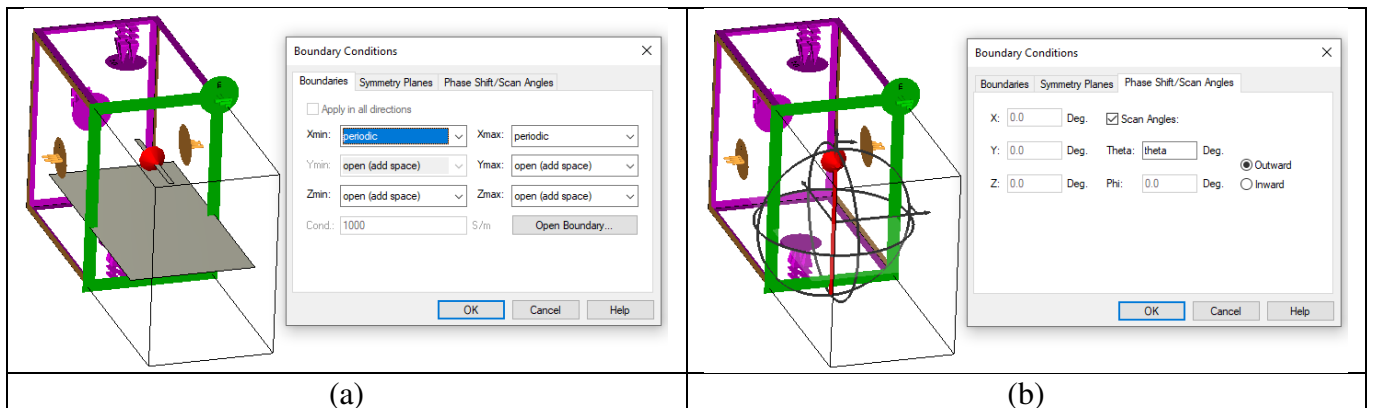


Figure 1 – Array element geometry.



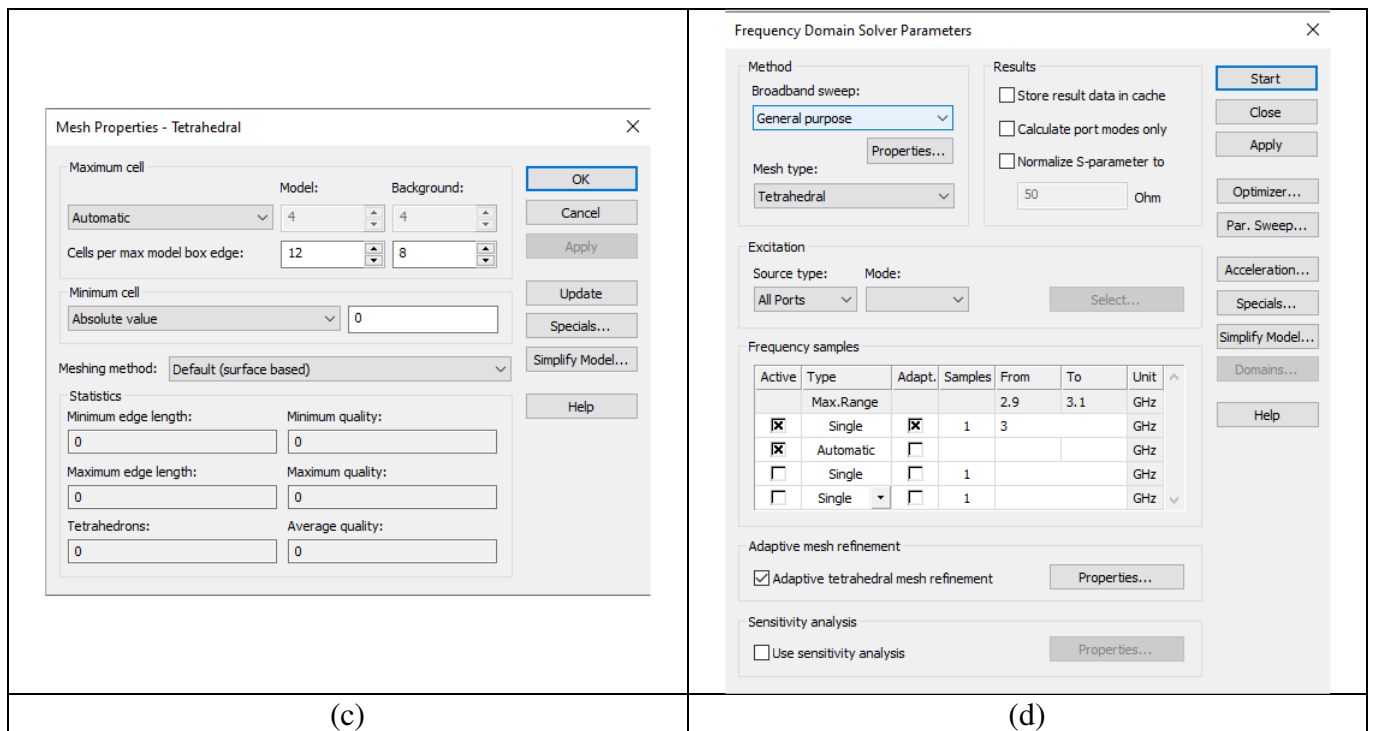


Figure 2 – Project settings.

Python Code

In this example the following applications are required:

1. CST Studio Suit 2020 [5].
2. Python 3.6 [6].
3. SciPy [7].

#load libraries for CST

```
import os
import sys
sys.path.append(r"C:\Program Files (x86)\CST STUDIO SUITE 2020\AMD64\python_cst_libraries")
import cst
import cst.interface
from cst.interface import Project
import cst.results
```

#load libraries for optimization

```
import numpy as np
from scipy.optimize import minimize
```

#define fitness function

```
def cst_fun(x):
```

#I want to use Nelder-Mead algorithm for optimization. However, it doesn't have bounds in #SciPy. So, I define the bounds by myself although it is a bad idea.

```
if 30.0<=x[0]<=70.0 and 20.0<=x[1]<=35.0:
```

```
    mycst = cst.interface.DesignEnvironment() #open CST
```

```
    mycst1=cst.interface.DesignEnvironment.open_project(mycst,
```

```
    r'C:\Optimizer\PhasedArray_1
```

```
    theta=0.cst') #open CST project, that will be optimized at 0deg.
```

```

#make VBA script to parameters change and project update
par_change = 'Sub Main () \nStoreParameter("dl", '+str(x[0])+')\nStoreParameter("h",
'+str(x[1])+')\nRebuildOnParametricChange (bfullRebuild, bShowErrorMsgBox)\nEnd Sub'
mycst1.schematic.execute_vba_code(par_change, timeout=None) #execute VBA script
cst.interface.DesignEnvironment.in_quiet_mode = True
mycst1.modeler.run_solver() #run simulation
cst.interface.DesignEnvironment.close(mycst) #close CST
#get S11
project = cst.results.ProjectFile(r'C:\Optimizer\PhasedArray_1 theta=0.cst')
S11 = project.get_3d().get_result_item(r"1D Results\S-Parameters\S1,1")
res=np.array(S11.get_ydata())
#find max of S11
opt_value1 = max(abs(res))

#repeat for the project, that will be optimized at 45deg.
mycst = cst.interface.DesignEnvironment()
mycst1=cst.interface.DesignEnvironment.open_project(mycst, r'C:\Optimizer\PhasedArray_1
theta=45.cst')
mycst1.schematic.execute_vba_code(h_change, timeout=None)
cst.interface.DesignEnvironment.in_quiet_mode = True
mycst1.modeler.run_solver()
cst.interface.DesignEnvironment.close(mycst)
project = cst.results.ProjectFile(r'C:\Optimizer\PhasedArray_1 theta=45.cst')
S11 = project.get_3d().get_result_item(r'1D Results\S-Parameters\S1,1')
res=np.array(S11.get_ydata())
opt_value2 = max(abs(res))

#I want to get minimum S11 at 0deg and 45deg, so I have to find max and then minimize it
print(str(opt_value1+opt_value2))
return opt_value1+opt_value2
#if we out of bounds
else:
    return 1.0

#set initial values of dl and h
x0 = np.array([35.0, 30.0])
#find minimum of the fitness function
res = minimize(cst_fun, x0, method='nelder-mead', options={'xatol': 1e-8, 'disp': True})

```

Results

The phased antenna array dimensions dl and h have been optimized at $\theta = 0^\circ$ (Fig. 3), $\theta = 45^\circ$ (Fig. 4), $\theta = 0^\circ$ and $\theta = 45^\circ$ (Fig. 5) using the presented script. It can be seen, that the maximum value of S_{11} is equal to -2.46 dB at $\theta = 45^\circ$ for the first case. The maximum value of S_{11} is equal to -4.1 dB at $\theta = 0^\circ$ for the second case. And the maximum value of S_{11} is equal to -4.85 dB at $\theta = 45^\circ$ for the third case.

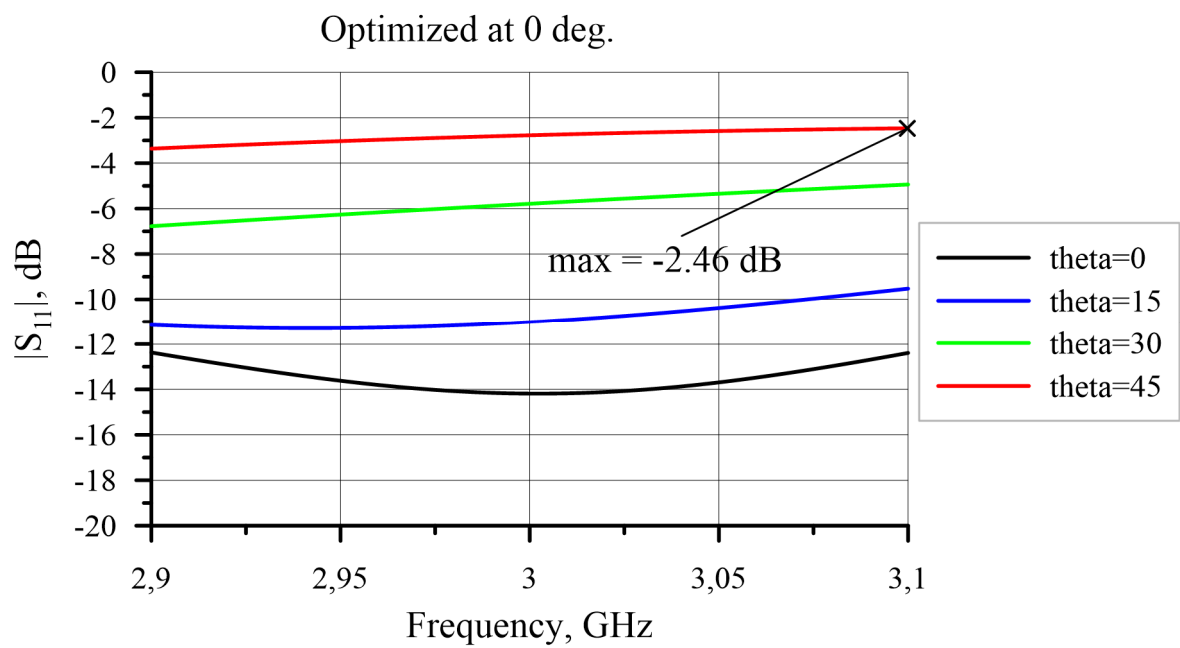


Figure 3 – S11 for different scanning angles optimized at $\theta = 0^\circ$.

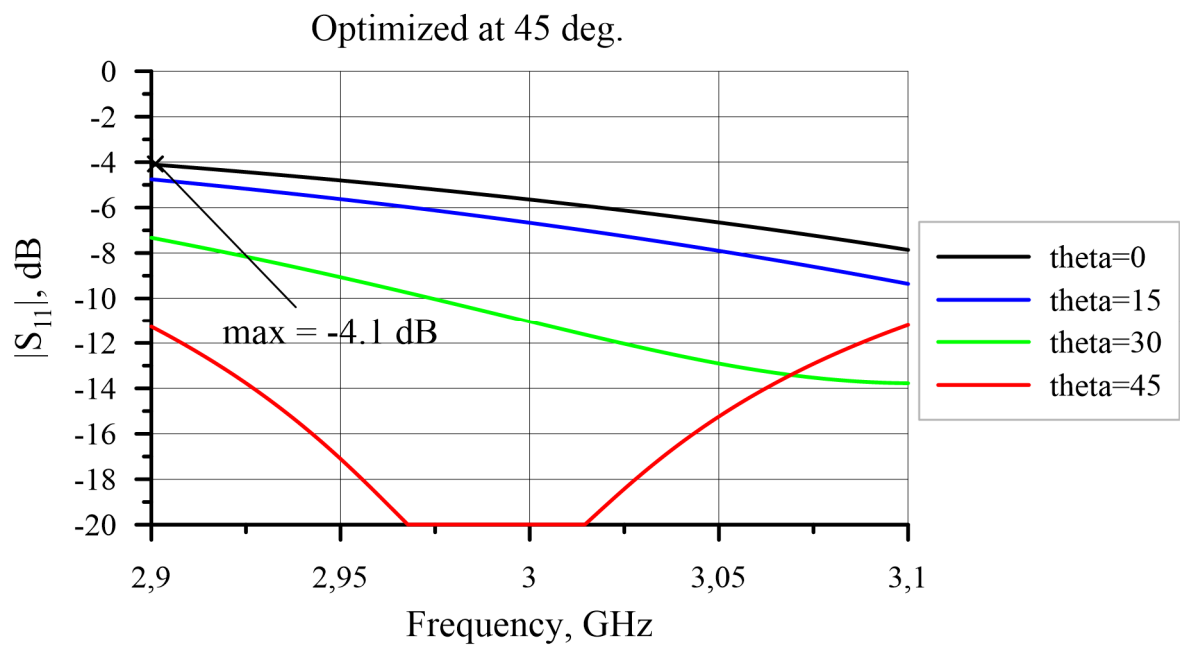


Figure 4 – S11 for different scanning angles optimized at $\theta = 45^\circ$.

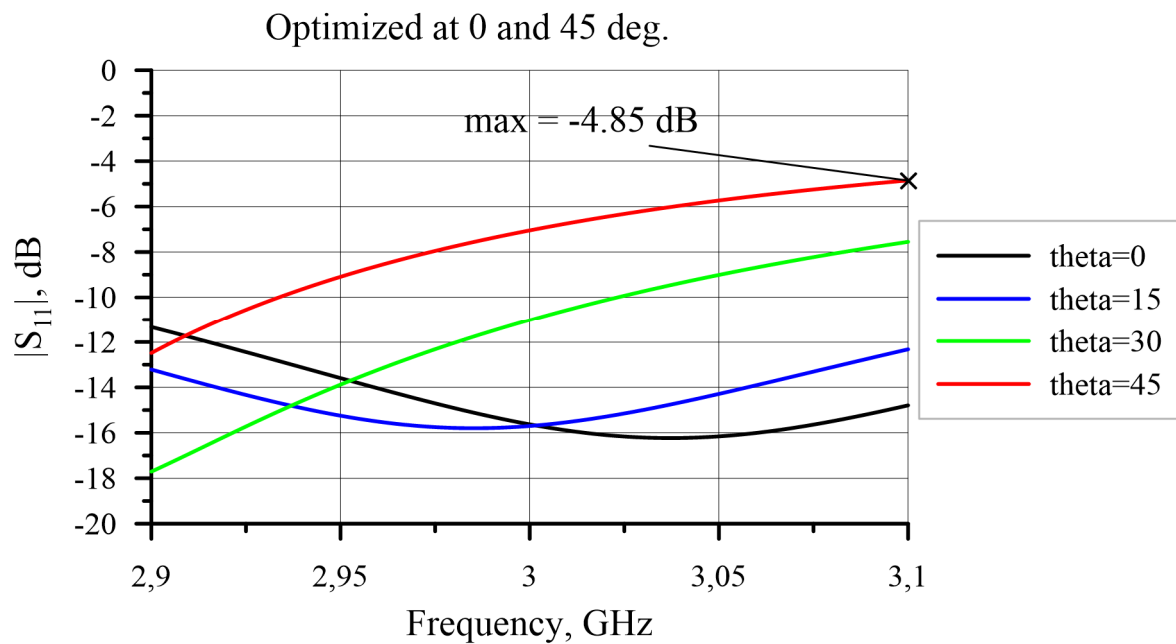


Figure 5 – S_{11} for different scanning angles optimized at $\theta = 0^\circ$ and $\theta = 45^\circ$.

Conclusion

In this text, an initial experience of using the Python cst package is presented. The python script, that allows to optimize phased antenna array is proposed. A simple example of using the script for dipole phased antenna array optimization is shown. By using the script, the S_{11} of the array has been reduced for 0.75 dB. The script code can be improved further. And it can be applied for optimization of real phased antenna arrays.

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7. <https://www.scipy.org/>